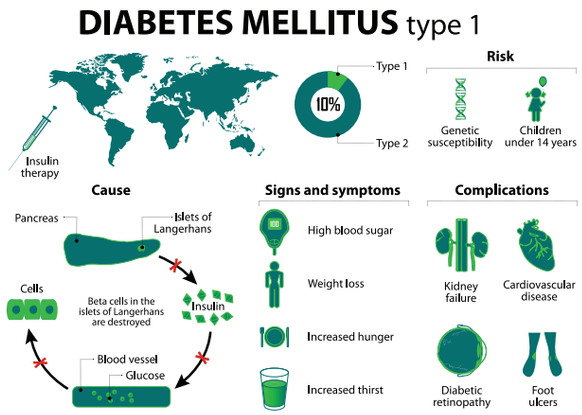
by **increased blood glucose level** due to **insufficient**, **deficient** or **failure of insulin secretion**. This is the most common pancreatic endocrine disorder. The incidence of Type-1 and Type-2 diabetes is increasing worldwide.

21.8.1 Type-1 Insulin Dependent Diabetes Mellitus (IDDM)

IDDM accounts for 10 to 20% of the known diabetics. The condition also occurs in children (**juvenile onset diabetes**) and **young adults**, the onset is usually sudden and can be life threatening. This is caused by the **destruction of β-cells of the pancreas**. It is characterized by abnormally elevated blood glucose levels (**hyperglycemia**) resulting from **inadequate insulin secretion**.

Causes: Genetic inheritance and environmental factors (infections due to virus, acute stress) are the cause for this condition.



21.8.2 Type-2 Non-Insulin Dependent Diabetes Mellitus (NIDDM)

This is also called as **adult onset diabetes** and accounting for 80 to 90% of the diabetic population. It develops slowly, usually milder and more stable. **Insulin production by the pancreas is normal** but its **action is impaired**. The target cells do not respond to insulin. It does not allow the movement of glucose into cells.

Causes: The causes are multifactorial which include increasing age (affecting middle aged and older people), obesity, sedentary life style, overeating and physically inactive.

More to Know

One in every 8 individuals in India is a diabetic. The revised WHO estimates for the year 2025 is 57.2 million diabetics in India. The average age for the onset of diabetes is 40 years, while it is 55 years in other countries. World Health Organizaation projects that diabetes will be 7th leading cause of death by the year 2030.

Symptoms: Diabetes mellitus is associated with several metabolic alterations.The most important symptoms are

- Increased blood glucose level (**Hyperglycemia**)
- Increased urine output (**Polyuria**) leading to dehydration
- Loss of water leads to thirst (**Polydipsia**) resulting in increased fluid intake
- Excessive glucose excreted in urine (**Glycosuria**)
- Excess hunger (**Polyphagia**) due to loss of glucose in urine.
- Fatigue and loss of weight



According to WHO recommendation, if the fasting blood glucose is greater than 140 mg/dl or the random blood glucose is greater than 200 mg /dl on more than two occasions, diagnosis for confirming diabetes is essential.

21.8.3 Prevention and Control of Diabetes

Diet, hypoglycemic drugs, insulin injection and exercise are the management options based on the type and severity of the condition. The overall goal of diabetes management is to maintain normal blood glucose level.

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